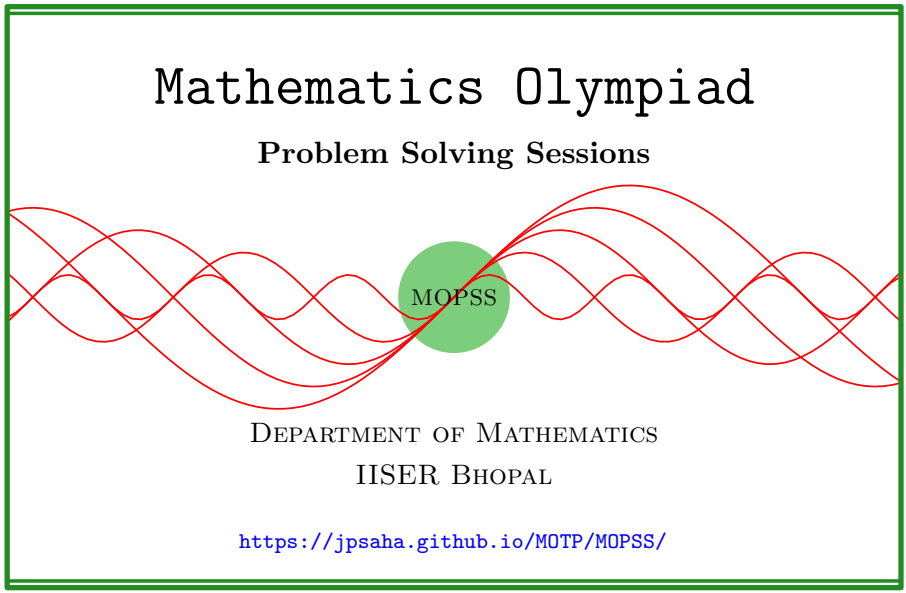
Invariance principle



Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

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§1 Invariants

See [**FGI96**, Chapter 12], [**Eng98**, Chapter 1].

§1.1 Impossibility

Exercise 1 (ELMO 1999 P2,). Mr. Fat moves around on the lattice points according to the following rules: From point (x, y) he may move to any of the points (y, x), $(3x, -2y)$, $(-2x, 3y)$, $(x+1, y+4)$ and $(x-1, y-4)$. Show that if he starts at $(0, 1)$ he can never get to $(0, 0)$.

Walkthrough — Show that under the given moves, the modulo 5 congruence class of the sum of the coordinates of the point where Mr. Fat is located remains invariant upto a factor of 3.

Solution 1. Note that if he moves from (x, y) to (a, b), then one of the congruences

$$a + b \equiv x + y \pmod 5, a + b \equiv 3(x + y) \pmod 5$$

hold. Since the sum of the coordinates of $(0, 1)$ is nonzero modulo 5, and the sum of the coordinates of $(0, 0)$ is zero, it follows that if he starts at $(0, 1)$ he can never get to $(0, 0)$. ■

Exercise 2 (RMO 2014b P6,). [Eng98, Problem 28, p. 29, p. 35] Let n be an odd positive integer and suppose that each square of an $n \times n$ grid is arbitrarily filled with either by 1 or by -1. Let r_j and c_k denote the product of all numbers in j-th row and k-th column respectively, $1 \leq j, k \leq n$. Prove that

$$\sum_{j=1}^{n} r_j + \sum_{k=1}^{n} c_k \neq 0.$$

Walkthrough —

(a) What would happen to the sum if the sign of one of the entries is changed? Does the sum change modulo 4?

(b) What would happen when all the entries are changed to 1 by changing the signs of the negative entries?

Solution 2. Denote the sum $\sum_{j=1}^{n} r_j + \sum_{k=1}^{n} c_k$ by S. If all entries of the grid are equal to 1, then S is equal to $2n$. Note that if the sign on an entry of the grid is changed, then S does not change modulo 4. So the integer S is congruent to $2n$ modulo 4. Since n is odd, the integer S is nonzero. For an alternate solution, see [Eng98, Problem 29, Chapter 2, p. 28]. ■

Exercise 3 (Cono Sur Olympiad 2013 P1,). Four distinct points are marked in a line. For each point, the sum of the distances from said point to the other three is calculated; getting in total 4 numbers. Decide whether these 4 numbers can be, in some order:

(i) $29, 29, 35, 37$,

(ii) $28, 29, 35, 37$,

(iii) $28, 34, 34, 37$.

Walkthrough —

(a) For each of the four points, express the sum of the distances to the other three points in terms of the distances between consecutive points among the four points.

(b) Show that at least two of the four numbers must be equal.

(c) How does it follow that the first set of numbers is the only one that can be obtained?

Solution 3. Denote the points by A, B, C, D from left to right. Let x, y, z be the distances between consecutive points, that is, $x = AB, y = BC, z = CD$. For P in $\{A, B, C, D\}$, let S_P be the sum of the distances from P to the other three points. Then, we have

$$\begin{aligned}
S_A &= AB + AC + AD \\
&= x + (x + y) + (x + y + z) \\
&= 3x + 2y + z, \\
S_B &= AB + BC + BD \\
&= x + y + (y + z) \\
&= x + 2y + z, \\
S_C &= AC + BC + CD \\
&= (x + y) + y + z \\
&= x + 2y + z, \\
S_D &= AD + BD + CD \\
&= (x + y + z) + (y + z) + z \\
&= x + 2y + 3z.
\end{aligned}$$

Note that $S_B = S_C$. Thus, at least two of the numbers S_A, S_B, S_C, S_D are equal. This is only the case for the first set of numbers, $29, 29, 35, 37$. Also observe that for $x = 3, y = 11, z = 4$, we have $S_A = 35, S_B = S_C = 29, S_D = 37$. This shows that only the first set of numbers can occur as the sums of the distances from the four points to the other three points in some order. ■

Exercise 4 (Japan Mathematical Olympiad 2018 P1,). Positive integers between 1 to 100 inclusive are written on a blackboard, each exactly once. One operation involves choosing two numbers a and b on the blackboard and erasing them, then writing the greatest common divisor of $a^2 + b^2 + 2$ and $a^2b^2 + 3$. After a number of operations, there is only one positive integer left on the blackboard. Prove this number cannot be a perfect square.

Walkthrough —

(a) Show that the parity of the number of multiples of 3 on the blackboard is an invariant.

(b) Show that the last number on the blackboard is a multiple of 3, but is not a multiple of 9.

Solution 4.

Claim — The parity of the number of multiples of 3 on the blackboard is an invariant.

Proof of the claim. Let a and b be the two numbers chosen in an operation. If a and b are both multiples of 3, then a^2+b^2+2 is not divisible by 3, and hence, $\gcd(a^2+b^2+2, a^2b^2+3)$ is not divisible by 3. If exactly one of a and b is a multiple of 3, then a^2+b^2+2 and a^2b^2+3 are both divisible by 3, and hence, $\gcd(a^2+b^2+2, a^2b^2+3)$ is divisible by 3. If neither a nor b is a multiple of 3, then a^2+b^2+2 is not divisible by 3, and hence, $\gcd(a^2+b^2+2, a^2b^2+3)$ is not divisible by 3. Therefore, the parity of the number of multiples of 3 on the blackboard remains unchanged after each operation. □

Initially, there are 33 multiples of 3 on the blackboard, so the parity of the number of multiples of 3 is odd. After performing 99 operations, there is only one number left on the blackboard, and it must be a multiple of 3. Let x, y be the last two numbers chosen in the final operation. Then, the last number on the blackboard is $\gcd(x^2+y^2+2, x^2y^2+3)$. This shows that 3 divides x^2y^2+3, which implies that 3 divides xy. Therefore, 9 does not divide x^2y^2+3, and hence, 9 does not divide $\gcd(x^2+y^2+2, x^2y^2+3)$. This shows that the last number on the blackboard is divisible by 3, but is not divisible by 9, and hence is not a perfect square. ■

Exercise 5 (Tournament of Towns 2021 Spring Junior A Level P2, proposed by Maxim Didin)**.** In a room there are several children and a pile of 1000 sweets. The children come to the pile one after another in some order. Upon reaching the pile each of them divides the current number of sweets in the pile by the number of children in the room, rounds the result if it is not integer, takes the resulting number of sweets from the pile and leaves the room. All the boys round upwards and all the girls round downwards. The process continues until everyone leaves the room. Prove that the total number of sweets received by the boys does not depend on the order in which the children reach the pile.

The following problem is similar to Exercise 9.

Exercise 6 (Singapore Junior Mathematical Olympiad 2019 Round 2 P2,)**.** There are 315 marbles divided into three piles of 81, 115 and 119. In

each move Ah Meng can either merge several piles into a single pile or divide a pile with an even number of marbles into 2 equal piles. Can Ah Meng divide the marbles into 315 piles, each with a single marble?

> **Walkthrough** —
> **(a)**

Solution 5. to write. ■

Exercise 7 (Belarus National Olympiad 2023 Grade 8 Day 1 P1,). A move on an unordered triple of numbers (a, b, c) changes the triple to either $(a, b, 2a + 2b - c)$, $(a, 2a + 2c - b, c)$ or $(2b + 2c - a, b, c)$. Can you perform a finite sequence of moves on the triple $(3, 5, 14)$ to get the triple $(3, 13, 6)$?

> **Walkthrough** — Show that if a sequence of moves is performed on a triple (a, b, c), then the resulting **unordered** triple is congruent to one of the triples
>
> $$(a, b, -(a + b + c)), (a, -(a + b + c), c), (-(a + b + c), b, c), (a, b, c)$$
>
> modulo 3, that is, the residues of the entries of the resulting triple modulo 3 coincide with the residues of the entries of one of the above triples in some order.

Solution 6. In the following, any triple is considered as an **unordered** triple.

> **Claim** — If a sequence of moves is performed on an **unordered** triple (a, b, c), then the resulting **unordered** triple is congruent to one of the **unordered** triples
>
> $$(a, b, -(a + b + c)), (a, -(a + b + c), c), (-(a + b + c), b, c), (a, b, c)$$
>
> modulo 3.

Proof of the Claim. It suffices to prove that an admissible move on any of the triples

$$(a, b, -(a + b + c)), (a, -(a + b + c), c), (-(a + b + c), b, c), (a, b, c)$$

results in a triple which is congruent to one of the above triples modulo 3. Note that applying an admissible move to the triple $(a, b, -(a + b + c))$ results in a triple which is congruent to one of the triples

$$(c, a, b), (c, a, -(a + b + c)), (c, b, -(a + b + c))$$

The content posted here and at this blog by Evan Chen are quite useful.

modulo 3. Similarly, applying an admissible move to the triple $(a, -(a+b+c), c)$ results in a triple which is congruent to one of the triples

$$(b, a, c), (b, a, -(a+b+c)), (b, c, -(a+b+c))$$

modulo 3. Applying an admissible move to the triple $(-(a+b+c), b, c)$ results in a triple which is congruent to one of the triples

$$(a, b, c), (a, b, -(a+b+c)), (a, c, -(a+b+c))$$

modulo 3. Finally, applying an admissible move to the triple (a, b, c) results in a triple which is congruent to one of the triples

$$(a, b, -(a+b+c)), (a, c, -(a+b+c)), (b, c, -(a+b+c))$$

modulo 3. In each case, the resulting triple is congruent to one of the triples

$$(a, b, -(a+b+c)), (a, -(a+b+c), c), (-(a+b+c), b, c), (a, b, c)$$

modulo 3. Thus, the claim is proved. □

Note that the triple $(3, 5, 14)$ is congruent to the triple $(0, 2, 2)$ modulo 3. By the claim, if a sequence of moves is performed on the triple $(3, 5, 14)$, then the resulting triple is congruent to one of the triples

$$(2, 2, 2), (0, 2, 2)$$

modulo 3. The triple $(3, 13, 6)$ is congruent to the triple $(0, 1, 0)$ modulo 3. Since $(0, 1, 0)$ is not congruent to either $(2, 2, 2)$ or $(0, 2, 2)$ modulo 3, it is impossible to perform a finite sequence of moves on the triple $(3, 5, 14)$ to get the triple $(3, 13, 6)$. ■

Remark. Note that $a^2 + b^2 + c^2 - ab - bc - ca$ is an invariant.

Exercise 8 (Belarus National Olympiad 2023 Grade 9 Day 1 P2,). A move on an unordered triple of numbers (a, b, c) changes the triple to either $(a, b, 2a + 2b - c)$, $(a, 2a + 2c - b, c)$ or $(2b + 2c - a, b, c)$. Can you perform a finite sequence of moves on the triple $(3, 5, 14)$ to get the triple $(9, 8, 11)$?

Walkthrough — Show that the mod 4 congruence class of the sum of the entries of such a triple remains invariant under the allowed moves if the sum of the entries of the initial triple is congruent to 2 modulo 4.

Solution 7. Note that for any unordered triple (a, b, c), the sum of its entries is congruent to the sum of the entries of any triple obtained from (a, b, c) by

an admissible move modulo 4 if the sum of the entries of the initial triple is congruent to 2 modulo 4. Indeed, the sum of the entries of any of the triples $(a, b, 2a + 2b - c)$, $(a, 2a + 2c - b, c)$ and $(2b + 2c - a, b, c)$ is congruent to $-(a + b + c)$ modulo 4. Hence, if the sum of the entries of the initial triple is congruent to 2 modulo 4, then the sum of the entries of any triple obtained from it by a finite sequence of moves is also congruent to 2 modulo 4. Since the sum of the entries of the triple $(3, 5, 14)$ is 22, which is congruent to 2 modulo 4, and the sum of the entries of the triple $(9, 8, 11)$ is 28, which is congruent to 0 modulo 4, it is impossible to perform a finite sequence of moves on the triple $(3, 5, 14)$ to get the triple $(9, 8, 11)$. ■

Remark. Note that $a^2 + b^2 + c^2 - ab - bc - ca$ is an invariant.

Exercise 9 (Kosovo National Olympiad 2023 Grade 11 P1,). In three different piles, there are 51, 49 and 5 stones, respectively. You can combine two piles in a larger pile or you can separate a pile that contains an even amount of stones into two equal piles. Is it possible that after some turns, we can separate the stones into 105 piles with 1 stone on each pile?

Walkthrough —

(a) Show that the greatest common divisor of the numbers of stones of the piles is an integer larger than 1.

(b) Show that it is impossible to separate the stones into 105 piles with 1 stone on each pile.

Solution 8. Note that if the greatest common divisor of the numbers of stones of the piles before some turn is an integer d, then d divides the number of stones in each pile after that turn. Since $5, 49, 51$ are all odd, it follows that during the first turn two piles are to be combined. Hence, the numbers of stones in the piles after the first turn are equal to $5, 100$, or to $49, 56$, or to $51, 54$. This shows that the greatest common divisor of the numbers of stones of the piles is an integer larger than 1. Consequently, it is impossible to separate the stones into 105 piles with 1 stone on each pile after some turns. ■

Exercise 10 (Canadian Mathematical Olympiad 2024 P2,). Jane writes down 2024 natural numbers around the perimeter of a circle. She wants the 2024 products of adjacent pairs of numbers to be exactly the set $\{1!, 2!, \ldots, 2024!\}$. Can she accomplish this?

Walkthrough —

(a) Show that to have such a configuration, the product $1! \cdot 2! \cdot \ldots \cdot 2024!$ must be a perfect square.

(b) Show that $1! \cdot 2! \cdot \ldots \cdot 2024!$ is not a perfect square.

Solution 9. We claim that Jane cannot accomplish this. Assume for the sake of contradiction that she can. Let the numbers be $a_1, a_2, \ldots, a_{2024}$ in clockwise order, and let $a_{2025} = a_1$. Then for some permutation $k_1, k_2, \ldots, k_{2024}$ of $1, 2, \ldots, 2024$, we have $a_i a_{i+1} = k_i!$ for any $1 \leq i \leq 2024$. It follows that

$$(a_1 a_2 a_3 \cdots a_{2024})^2 = 1! \cdot 2! \cdot \ldots \cdot 2024!.$$

Note that 1009 divides each of $1009!, 1010!, \ldots, 2024!$ but does not divide any of $1!, 2!, \ldots, 1008!$. Moreover, 1009^2 does not divide any of $1009!, 1010!, \ldots, 2017!$. Also note that 1009^2 divides $2018!, 2019!, \ldots, 2024!$, and 1009^3 does not divide any of $2018!, 2019!, \ldots, 2024!$. This implies that the largest nonnegative integer r such that 1009^r divides $1! \cdot 2! \cdot \ldots \cdot 2024!$ is equal to

$$(2024 - 1008) + (2024 - 2017),$$

which is odd. This implies that $1! \cdot 2! \cdot \ldots \cdot 2024!$ is not a perfect square, contradicting the earlier equation. Hence, Jane cannot accomplish this. ■

Remark. Note that no prime larger than 1009 helps to show that $1! \cdot 2! \cdot \ldots \cdot 2024!$ is not a perfect square.

§1.2 Determining certain quantities

Example 11. Suppose the numbers $1, \frac{1}{2}, \frac{1}{3}, \ldots, \frac{1}{100}$ are written on a blackboard. In each step, you may erase two numbers a and b and replace them with the number $a + b + ab$. After 99 steps, only one number remains. What are the possible values of this number?

Walkthrough —

(a)

Solution 10. For any two rationals a and b, define

$$a \oplus b = a + b + ab = (a + 1)(b + 1) - 1,$$

and call $a \oplus b$ the *nice sum* of a and b. Note that the operation $\oplus$ is associative and commutative. Hence, we may consider the nice sum of all the numbers on the blackboard, which is well-defined. The nice sum of a single number a is defined as $a \oplus 0 = a$. Note that the nice sum of the numbers present on the blackboard at a given time does not change after a step, since the two numbers

a and b are replaced by their nice sum $a \oplus b$. Initially, the nice sum of the numbers on the blackboard is

$$(1+1)\left(\frac{1}{2}+1\right)\left(\frac{1}{3}+1\right)\cdots\left(\frac{1}{100}+1\right)-1=101-1=100.$$

Therefore, the nice sum of the numbers on the blackboard is always 100. In particular, the nice sum of the final number is 100, so the final number is 100 regardless of the choices made in the steps. ■

Exercise 12 (Argentina National Olympiad 2018 Level 2 P1,). A list of 2018 numbers is created using the following procedure: the first number is 47, the second number is 74, and from there, each number is equal to the number formed by the last two digits of the sum of the two previous numbers:

$$47, 74, 21, 95, 16, 11, \ldots$$

Bruno squares each of the 2018 numbers and sums them. Determine the remainder when this sum is divided by 8.

Walkthrough —

(a) Consider the residues of the numbers in the list modulo 4.

(b) Determine the residues of the squares of the numbers in the list modulo 8.

(c) Determine the remainder when the sum of the squares of the numbers in the list is divided by 8.

Solution 11. Note that the first two integers on the list are congruent to 3 and 2 modulo 4, respectively. Therefore, the next six integers on the list are congruent to 1, 3, 0, 3, 3, and 2 modulo 4, respectively. Applying induction, it follows that the residues of the numbers in the list modulo 4 are periodic with period 6:

$$3, 2, 1, 3, 0, 3, 3, 2, 1, 3, 0, 3, \ldots.$$

Hence, the residues of the squares of the numbers in the list modulo 8 are also periodic with period 6:

$$1, 4, 1, 1, 0, 1, 1, 4, 1, 1, 0, 1, \ldots.$$

Since $2018 = 6 \cdot 336 + 2$, the sum of the squares of the numbers in the list is congruent to

$$336 \cdot (1+4+1+1+0+1)+1+4$$

modulo 8. Therefore, the remainder when this sum is divided by 8 is 5. ■

Exercise 13 (Balkan Mathematical Olympiad 2018 P2, , proposed by Jeremy King, UK)**.** Let q be a positive rational number. Two ants are initially at the same point X in the plane. In the n-th minute ($n = 1, 2, \dots$) each of them chooses whether to walk due north, east, south or west and then walks the distance of q^n metres. After a whole number of minutes, they are at the same point in the plane (not necessarily X), but have not taken exactly the same route within that time. Determine all possible values of q.

Walkthrough —

(a) Let q be a positive rational number such that the two ants can be at the same point in the plane after a whole number of minutes, without taking the same route.

(b) Let r be a positive integer such that the two ants are at the same point in the plane after r minutes, without taking the same route.

(c) Prove that

$$\varepsilon_1 q + \varepsilon_2 q^2 + \cdots + \varepsilon_r q^r = \varepsilon_1' q + \varepsilon_2' q^2 + \cdots + \varepsilon_r' q^r,$$

for some $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_r$, $\varepsilon_1', \varepsilon_2', \dots, \varepsilon_r'$ are elements of $\{1, -1, i, -i\}$, and the tuples $(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_r)$ and $(\varepsilon_1', \varepsilon_2', \dots, \varepsilon_r')$ are not equal.

(d) Prove that there exists a nonzero polynomial P of degree at most $r - 1$, which vanishes at q, and has coefficients in $\{0, 2, 1 + i\}$ up to a factor of $\pm 1, \pm i$.

(e) Further, prove that there exists a nonzero polynomial Q of degree at most $r - 1$, which vanishes at q, and has coefficients in $\{0, 1, 1 + i\}$ up to a factor of $\pm 1, \pm i$.

(f) Write $q = \frac{a}{b}$ where a, b are relatively prime positive integers, and show that b divides the leading coefficient of Q in $\mathbb{Z}[i]$, and a divides the constant term of Q in $\mathbb{Z}[i]$, where

$$\mathbb{Z}[i] = \{x + yi : x, y \in \mathbb{Z}\}$$

is the ring of Gaussian integers.

(g) Prove further that a divides the constant term of Q in $\mathbb{Z}[i]$.

(h) Conclude that $q = 1$ is the only possible value of q.

Solution 12. Let (a, b) denote the coordinates of the point X. Let (x_n, y_n) and (x_n', y_n') be the coordinates of the two ants after n minutes. We have $x_0 = x_0' = a$ and $y_0 = y_0' = b$. Then for each n, note that $(x_n, y_n) - (x_{n-1}, y_{n-1})$ is equal to one of $(0, q^n), (q^n, 0), (0, -q^n), (-q^n, 0)$. Similarly, $(x_n', y_n') - (x_{n-1}', y_{n-1}')$ is equal to one of $(0, q^n), (q^n, 0), (0, -q^n), (-q^n, 0)$. Let r be a positive integer such that $(x_r, y_r) = (x_r', y_r')$. Then we have

$$(a, b) + \sum_{n=1}^{r} (x_n - x_{n-1}, y_n - y_{n-1}) = (a, b) + \sum_{n=1}^{r} (x_n' - x_{n-1}', y_n' - y_{n-1}'),$$

which implies that

$$\sum_{n=1}^{r}(x_n - x_{n-1}, y_n - y_{n-1}) = \sum_{n=1}^{r}(x'_n - x'_{n-1}, y'_n - y'_{n-1}).$$

This shows that

$$\sum_{n=1}^{r}((x_n + iy_n) - (x_{n-1} + iy_{n-1})) = \sum_{n=1}^{r}\left((x'_n + iy'_n) - (x'_{n-1} + iy'_{n-1})\right).$$

For $1 \leq n \leq r$, write

$$(x_n + iy_n) - (x_{n-1} + iy_{n-1}) = q^n i^{k_n}$$

and

$$(x'_n + iy'_n) - (x'_{n-1} + iy'_{n-1}) = q^n i^{k'_n},$$

where k_n and k'_n are integers. Also observe that k_n and k'_n are not congruent modulo 4 for some n with $1 \leq n \leq r$, since the two ants do not take the same route within r minutes. Then we have

$$\sum_{n=1}^{r} q^n i^{k_n} = \sum_{n=1}^{r} q^n i^{k'_n},$$

which implies that

$$\sum_{n=1}^{r} q^{n-1}(i^{k_n} - i^{k'_n}) = 0.$$

Observing

$$1 - i = -i(1 + i),$$

it follows that for each n with $1 \leq n \leq r$, the difference $i^{k_n} - i^{k'_n}$ is equal to $0, 2, 1 + i$ up to a factor of $\pm 1, \pm i$. Hence, there exists a nonzero polynomial P of degree at most $r - 1$, which vanishes at q, and has coefficients in $\{0, 2, 1 + i\}$ up to a factor of $\pm 1, \pm i$. Since q is nonzero, it follows that P is a nonconstant polynomial, and hence, we may (and do) assume that P has a nonzero constant term. Further, dividing the coefficients of P by $1 + i$, we may (and do) assume that P has coefficients in $\{0, 1, 1 + i\}$ up to a factor of $\pm 1, \pm i$. Write $q = \frac{a}{b}$ for some relatively prime positive integers a, b. Considering the real and imaginary parts of the coefficients of P separately, and applying the rational root theorem to the resulting polynomials with integer coefficients, it follows that a divides the constant term of P in $\mathbb{Z}[i]$, and b divides the leading coefficient of P in $\mathbb{Z}[i]$. This implies that the square of the absolute value of a divides the square of the absolute value of the constant term of P, and the square of the absolute value of b divides the square of the absolute value of the leading coefficient of P. It follows that the square of the absolute value of each of a and b is at most 2, and hence $a = b = 1$. This shows that $q = 1$ is the only possible value of q.

Finally, we verify that $q = 1$ indeed works. Let the first ant walk north, east in the first two minutes, and the second ant walk east, north in the first two minutes. Then they are at the same point after two minutes, but have not taken the same route within that time. ■

Exercise 14 (Brazil National Olympiad 2018 Level 2 P5,). Initially, the numbers $1, 2, 3, \ldots, 10$ are written on a board. An operation consists of choosing two numbers a and b from the board, deleting them, and writing the number $a+b+\frac{ab}{f(a,b)}$, where $f(a, b)$ denotes the sum of all numbers currently on the board excluding a and b. This process is repeated until only two numbers x and y remain, with $x \geq y$. What is the maximum possible value of x?

> **Walkthrough** — Show that the sum of the product of the pairs of the numbers written on the board is an invariant.

Solution 13. Let S denote the sum of the product of the pairs of the numbers, present on the board, prior to an operation. Let a and b be the numbers chosen for the operation. Let s denote the sum of the numbers prior to the operation, excluding a and b, and let T denote the sum of the product of the pairs of the numbers on the board, after the operation. Note that

$$\begin{aligned} T &= S - ab - as - bs + \left(a + b + \frac{ab}{s}\right) s \\ &= S. \end{aligned}$$

This shows that the sum of the product of the pairs of the numbers on the board is an invariant. Initially, the numbers $1, 2, 3, \ldots, 10$ are on the board, and thus

$$\begin{aligned} S &= \sum_{1 \leq i < j \leq 10} ij \\ &= \frac{1}{2}\left((1 + 2 + \cdots + 10)^2 - (1^2 + 2^2 + \cdots + 10^2)\right) \\ &= \frac{1}{2}\left(55^2 - 385\right) \\ &= 1320. \end{aligned}$$

This shows that the sum of the product of the pairs of the numbers on the board is always 1320. Finally, when only two numbers x and y remain, we have $xy = 1320$. This implies that $x \leq 1320$.

Now we show that the maximum possible value of x is 1320, that is, we show that it is possible to perform the operations in such a way that the last two numbers on the board are 1320 and 1. Indeed, if we exclude the number 1 in the first eight operations, then the last two numbers on the board are 1320 and 1. This shows that the maximum possible value of x is 1320. ■

Exercise 15 (Brazil National Olympiad 2022 Level 3 P1,). A single player game has the following rules: initially, there are 10 piles of stones with $1, 2, \ldots, 10$ stones, respectively. A movement consists of making one of the following operations:

(i) to choose 2 piles, both of them with at least 2 stones, combine them and then add 2 stones to the new pile;

(ii) to choose a pile with at least 4 stones, remove 2 stones from it, and then split it into two piles with amount of stones in those two piles to be chosen by the player.

The game continues until it is not possible to make an operation. Show that the number of piles with one stone in the end of the game is always the same, no matter how the movements are made.

Walkthrough —

(a) Let p denote the number of piles and s denote the total number of stones at some stage of the game.

(b) Note that after applying operation (i), (p, s) changes to $(p-1, s+2)$, and after applying operation (ii), (p, s) changes to $(p+1, s-2)$.

(c) Construct a polynomial in p and s using the above observations, which remains invariant under both the operations.

Solution 14. Let p denote the number of piles and s denote the total number of stones at some stage of the game. Note that after applying operation (i), we have $p-1$ piles and $s+2$ stones. Moreover, after applying operation (ii), we have $p+1$ piles and $s-2$ stones. This shows that $2p+s$ remains invariant under both the operations. Let P denote the number of piles when the game has come to an end by performing the operations in a certain sequence. Note that there are at least $P-1$ piles containing only one stone. Let S denote the number of stones in the remaining pile. Observe that $1 \leq S \leq 3$. The total number of stones is equal to $S + (P-1)$. This yields

$$2P + S + P - 1 = 2 \times 10 + (1 + 2 + \cdots + 10),$$

which implies

$$3P + S = 2 \times 10 + (1 + 2 + \cdots + 10) + 1 = 20 + 55 + 1 = 76.$$

This implies that $P = 25$ and $S = 1$, so there are 25 piles with one stone in the end of the game. ■

Remark. Note that applying the second operation to the pile obtained by applying the first operation, may yield the piles, to which the first operation was applied. Thus the game need not terminate. This problem seems to ask that the number of piles with one stone in the end of the game (when an end state is reached) is always the same, no matter how the movements are made.

Exercise 16 (Belarus National Olympiad 2022 Grade 9 Day 1 P4,). The numbers $1, 2, \ldots, 50$ are written on the board. Anya does the following operation: removes the numbers a and b from the board and writes their sum $a + b$, after which she also notes down the number $ab(a + b)$. After 49 of these operations, only one number is left on the board. Anya summed up all the 49 numbers in her notes and got S.

(a) Prove that S does not depend on the order of Anya's actions.

(b) Calculate S.

Walkthrough —

(a) Let us denote the sum of the cubes of the numbers on the board at a moment as B, and the sum of the numbers in Anya's notes at the same moment as N.

(b) Prove that $B - 3N$ is invariant under Anya's operation.

(c) Calculate S using the invariance of $B - 3N$.

Solution 15. Let B denote the sum of the cubes of the numbers on the board at a moment, and N denote the sum of the numbers in Anya's notes at the same moment. We will show that $B - 3N$ is invariant under Anya's operation. Let a and b be the numbers removed from the board. Note that B changes to $B - a^3 - b^3 + (a + b)^3$ and N changes to $N + ab(a + b)$. This shows that $B - 3N$ changes to

$$B - a^3 - b^3 + (a + b)^3 - 3(N + ab(a + b)) = B - 3N.$$

Since $B - 3N$ is invariant, we have

$$1^3 + 2^3 + \cdots + 50^3 = (1 + 2 + \cdots + 50)^3 - 3S,$$

which gives

$$\begin{aligned} S &= \frac{(1 + 2 + \cdots + 50)^3 - (1^3 + 2^3 + \cdots + 50^3)}{3} \\ &= \frac{(1 + 2 + \cdots + 50)^3 - (1 + 2 + \cdots + 50)^2}{3} \\ &= \frac{1}{3}\left(\left(\frac{50 \cdot 51}{2}\right)^2 \cdot \left(\frac{50 \cdot 51}{2} - 1\right)\right) \end{aligned}$$

$$= \frac{1275^2 \cdot 1274}{3}.$$

■

Exercise 17 (British Mathematical Olympiad Round 2 2023 P2)**.** For an integer $n > 1$, the numbers $1, 2, 3, \ldots, n$ are written in order on a blackboard. The following *moves* are possible:

(i) Take three adjacent numbers x, y, z whose sum is a multiple of 3 and replace them with y, z, x.

(ii) Take two adjacent numbers x, y whose difference is a multiple of 3 and replace them with y, x.

For example we could take: $1, 2, 3, 4 \xrightarrow{\text{(i)}} 2, 3, 1, 4 \xrightarrow{\text{(ii)}} 2, 3, 4, 1$. Find all n such that the initial list can be transformed into $n, 1, 2, \ldots, n-1$ after a finite number of moves.

Walkthrough —

(a)

Solution 16. Let $x_1, x_2, \ldots, x_n$ denote the numbers on the blackboard, in the order they are written, prior to a move. We will show that the modulo 3 congruence class of the following quantity is invariant under the moves:

$$S = \sum_{i=1}^{n} i x_i.$$

For move (i), we have $x_i = x$, $x_{i+1} = y$ and $x_{i+2} = z$ for some $1 \leq i \leq n-2$. Then the change in S is

$$\begin{aligned}
&(iy + (i+1)z + (i+2)x) - (ix + (i+1)y + (i+2)z) \\
&= -y - z + 2x \\
&\equiv -(x + y + z) \pmod 3 \\
&\equiv 0 \pmod 3.
\end{aligned}$$

For move (ii), we have $x_i = x$ and $x_{i+1} = y$ for some $1 \leq i \leq n-1$. Then the change in S is

$$iy + (i+1)x - (ix + (i+1)y) = x - y \equiv 0 \pmod 3.$$

Thus S is invariant modulo 3. Initially, we have

$$S = 1 \cdot 1 + 2 \cdot 2 + \cdots + n \cdot n$$

$$= 1^2 + 2^2 + \cdots + n^2.$$

Let $n > 1$ be an integer such that the initial list can be transformed into $n, 1, 2, \ldots, n-1$ after a finite number of moves. Then we also have

$$\begin{aligned} S &= 1 \cdot n + 2 \cdot 1 + 3 \cdot 2 + \cdots + n \cdot (n-1) \\ &= n + (2^2 - 2) + (3^2 - 3) + \cdots + (n^2 - n) \\ &\equiv 1^2 + 2^2 + \cdots + n^2 - (1 + 2 + \cdots + n) + n. \end{aligned}$$

Since S is invariant modulo 3, we have

$$1^2 + 2^2 + \cdots + n^2 \equiv 1^2 + 2^2 + \cdots + n^2 - (1 + 2 + \cdots + n) + n \pmod 3,$$

which simplifies to

$$1 + 2 + \cdots + n \equiv n \pmod 3.$$

This gives

$$\frac{n(n-1)}{2} \equiv 0 \pmod 3,$$

and hence, either $n \equiv 0 \pmod 3$ or $n \equiv 1 \pmod 3$.

Claim — Let $n > 1$ be an interger satisfying $n \equiv 0 \pmod 3$ or $n \equiv 1 \pmod 3$. Then the initial list can be transformed into $n, 1, 2, \ldots, n-1$ after a finite number of moves.

Proof of the claim. Note that the move (i) can be applied to $1, 2, 3$ to get $2, 3, 1$, and the move (i) can be applied again to $2, 3, 1$ to get $3, 1, 2$. Thus we can transform $1, 2, 3$ into $3, 1, 2$. Similarly, the move (i) can be applied to $1, 2, 3, 4$ to get $1, 3, 4, 2$, and the move (i) can be applied again to $1, 3, 4, 2$ to obtain $1, 4, 2, 3$. Applying the move (ii) to $1, 4, 2, 3$ gives $4, 1, 2, 3$. Thus we can transform $1, 2, 3, 4$ into $4, 1, 2, 3$. Let k be a positive integer such that the list $1, 2, \ldots, n$ can be transformed into $n, 1, 2, \ldots, n-1$ for all $1 < n \leq 3k + 1$ such that 3 divides $n(n-1)$.

Consider the list $1, 2, \ldots, 3k + 3$. Since $4, 5, \ldots, 3k + 3$ are congruent to $1, 2, \ldots, 3k$ modulo 3 respectively, by the induction hypothesis, we can transform $4, 5, \ldots, 3k+3$ into $3k+3, 4, 5, \ldots, 3k+2$. Applying the move (ii) to $1, 2, 3, 3k+3, 4, 5, \ldots, 3k + 2$ gives $1, 2, 3k + 3, 3, 4, 5, \ldots, 3k + 2$. Applying the move (i) to $1, 2, 3k + 3$ twice gives $3k + 3, 1, 2$. Thus we can transform $1, 2, \ldots, 3k + 3$ into $3k + 3, 1, 2, 3, 4, \ldots, 3k + 2$.

Consider the list $1, 2, \ldots, 3k + 4$. Since $4, 5, \ldots, 3k + 4$ are congruent to $1, 2, \ldots, 3k + 1$ modulo 3 respectively, by the induction hypothesis, we can transform $4, 5, \ldots, 3k + 4$ into $3k + 4, 4, 5, \ldots, 3k + 3$. Applying move (i) to $1, 2, 3, 3k + 4$ gives $1, 2, 3k + 4, 3$. Applying move (i) to $1, 2, 3k + 4$ twice gives $3k+4, 1, 2$. Thus we can transform $1, 2, \ldots, 3k+4$ into $3k+4, 1, 2, 3, 4, \ldots, 3k+3$.

This completes the induction step, and hence the claim holds by induction. □

■

Exercise 18 (Chip-firing game)**.** Assume that on the Cartesian plane, four chips are placed at the origin. In each step, you may choose a lattice point (x, y) having at least a chip on it, remove one chip from (x, y), and place one chip each at $(x+1, y)$ and one at $(x, y+1)$. Show that after a finite number of steps, there are at least two chips at some lattice point.

Remark. The problem seems to ask that after any number of finitely many moves, there will exist two coins placed at the same point.

Walkthrough —

(a)

Solution 17. Let S denote the sum of $n_{(x,y)}r^{x+y}$ over all points (x, y), where there is a chip at (x, y) at some stage of the process, and $n_{(x,y)}$ denotes the number of chips at (x, y) at that stage, and r is a real number to be determined later. If we remove a chip at (x, y) and place one chip each at $(x+1, y)$ and $(x, y+1)$, then S changes to

$$S - r^{x+y} + r^{x+1+y} + r^{x+y+1} = S + r^{x+y}(2r - 1).$$

If we choose $r = \frac{1}{2}$, then S does not change after any move. Initially, $S = 4$ since there are four chips at the origin.

If there are no two chips at the same point at some stage of the process after finitely many moves, then S is the sum of r^{x+y} over all points (x, y) where there is a chip, and this implies that

$$\begin{aligned} 4 &= S \\ &< \sum_{(x,y)\in\mathbb{Z}^2_{\geq 0}} r^{x+y} \\ &= \sum_{x=0}^{\infty}\sum_{y=0}^{\infty} r^{x+y} \\ &= \left(\sum_{x=0}^{\infty} r^x\right)^2 \\ &= \frac{1}{(1-r)^2} \\ &= 4, \end{aligned}$$

which is a contradiction. Thus, there are at least two chips at some lattice point after a finite number of steps. ■

Exercise 19 (Serbia IMO TST 2024 Day 1 P1,). Three coins are placed at the origin of a Cartesian coordinate system. On one move one removes a coin placed at some position (x, y) and places three new coins at $(x+1, y)$, $(x, y+1)$ and $(x+1, y+1)$. Prove that after finitely many moves, there will exist two coins placed at the same point.

Remark. The problem seems to ask that after any number of finitely many moves, there will exist two coins placed at the same point.

Walkthrough —

(a)

Solution 18. Let S denote the sum of $n_{(x,y)}r^{x+y}$ over all points (x, y), where there is a coin at (x, y) at some stage of the process, and $n_{(x,y)}$ denotes the number of coins at (x, y) at that stage, and r is a real number to be determined later. If we remove a coin at (x, y) and place three coins at $(x+1, y)$, $(x, y+1)$ and $(x+1, y+1)$, then S changes to

$$S - r^{x+y} + r^{x+1+y} + r^{x+y+1} + r^{x+1+y+1} = S + r^{x+y}(2r + r^2 - 1).$$

If we choose r such that $2r + r^2 - 1 = 0$, then S does not change after any move. Let us take $r = \sqrt{2} - 1$. Since it is root of the equation $2r + r^2 - 1 = 0$, it follows that S is invariant under the moves. Initially, $S = 3$ since there are three coins at the origin.

If there are no two coins at the same point at some stage of the process after finitely many moves, then S is the sum of r^{x+y} over all points (x, y) where there is a coin, and this implies that

$$\begin{aligned}
3 = S &< \sum_{(x,y)\in\mathbb{Z}^2_{\geq 0}} r^{x+y} \\
&= \sum_{x=0}^{\infty}\sum_{y=0}^{\infty} r^{x+y} \\
&= \left(\sum_{x=0}^{\infty} r^x\right)^2 \\
&= \frac{1}{(1-r)^2} \\
&= \left(1 + \frac{1}{\sqrt{2}}\right)^2 \\
&= \frac{3}{2} + \sqrt{2}
\end{aligned}$$

$$< 3,$$

which is a contradiction. This proves that after any number of finitely many moves, there will exist two coins placed at the same point. ■

Exercise 20 (St Petersburg Mathematical Olympiad 2025 Round 2 Grade 11 P1,). 100 red and 99 blue piranhas were released into the empty Quiet Pool. When a piranha wants to eat, it can eat two other piranhas. If its victims are of the same color, the piranha changes its color, but if they are multicolored, it does not change. In the end, only one piranha remains. What color is it?

Walkthrough —

(a) Show that the number of blue piranhas and the number of red piranhas differ by a multiple of 4.

(b) Show that a red piranha remains at the end of the process.

Solution 19. Let r, b denote the number of red and blue piranhas, respectively, before a piranha eats. If a red piranha eats two red piranhas, then r, b changes to $r-3, b+1$ respectively. If a red piranha eats two blue piranhas, then r, b changes to $r-1, b-1$ respectively. If a blue piranha eats two red piranhas, then r, b changes to $r-1, b-1$ respectively. If a blue piranha eats two blue piranhas, then r, b changes to $r+1, b-3$ respectively. If the victims are of different colors, then r, b changes to $r-1, b-1$ respectively, regardless of the color of the eater. If R (resp. B) denotes the number of red (resp. blue) piranhas at the end of this eating, then the integers $R-B, r-b$ differ by a multiple of 4. This shows that the difference between the number of red and blue piranhas remains the same modulo 4 throughout the process. Since initially there are 100 red and 99 blue piranhas, the difference is $100-99=1$. Hence, the difference between the number of red and blue piranhas at the end is also 1 (mod 4). Therefore, the last remaining piranha must be red, since if it were blue, the difference would be -1, which is not congruent to 1 (mod 4). ■

Remark. We can ask to determine the colors of the remaining piranhas at the end, if initially there were 100 red and 100 blue piranhas.

Exercise 21 (Dutch Mathematical Olympiad 2025 P3,). Amber has a square consisting of 10×10 square boxes. One at a time, she randomly chooses one of the hundred squares and fills it with a number. The number she puts in a square is always equal to the total number of squares already filled in that square's row and column. So in the first square she chooses she writes a 0, in the second square a 0 or a 1, depending on whether it is in a different row and a different column or not, et cetera. So in each square there will be a number from 0 to 18 inclusive. After Amber has written a number in each square, she adds up all the numbers. What is/are the possible result(s) of this addition?

Walkthrough —

(a) In a chosen square, instead of writing the sum of the number of filled squares in the row and column, let us write down the pair of numbers (r, c) where r is the number of filled squares in that row before filling the square, and c is the number of filled squares in that column before filling the square.

(b) It amounts to find the sum of the numbers of the form $r + c$ across all the squares.

(c) Try to sum up the r values across all the squares, and then sum up the c values across all the squares.

(d) Show that the result of the addition is always the same, regardless of the order in which Amber fills the squares.

Solution 20. Let us allow Bob to play along with Amber. As Amber fills a square, Bob fills the same square with a pair (r, c) where r is the number of filled squares in that row before Amber fills it, and c is the number of filled squares in that column before Amber fills it. Note that the number Amber fills in that square is $r + c$. After Amber fills all the squares, Bob will have filled all the squares with pairs of numbers. Let us denote the sum of all the numbers Amber filled in as S. We have

$$S = \sum_{(r,c)} (r + c) = \sum_{(r,c)} r + \sum_{(r,c)} c.$$

Now, let us look at the first sum $\sum_{(r,c)} r$. For each row, the value of r across the squares in that row is $0, 1, 2, \ldots, 9$ in some order. Therefore, the sum of r across the squares in that row is $0 + 1 + 2 + \cdots + 9 = 45$. Since there are 10 rows, we have

$$\sum_{(r,c)} r = 10 \cdot 45 = 450.$$

By a similar argument, we have

$$\sum_{(r,c)} c = 10 \cdot 45 = 450.$$

Therefore, we have

$$S = \sum_{(r,c)} r + \sum_{(r,c)} c = 450 + 450 = 900.$$

Thus, the only possible result of the addition is 900. ■

Exercise 22 (RMO 2016d P4,). A box contains 4032 answer scripts out of which exactly half have odd number of marks. We choose 2 scripts randomly

and, if the scores on both of them are odd number, we add one mark to one of them, put the script back in the box and keep the other script outside. If both scripts have even scores, we put back one of the scripts and keep the other outside. If there is one script with even score and the other with odd score, we put back the script with the odd score and keep the other script outside. After following this procedure a number of times, there are 3 scripts left among which there is at least one script each with odd and even scores. Find, with proof, the number of scripts with odd scores among the three left.

Walkthrough — What happens to the number of scripts with odd scores?

Solution 21. Note that under this process, the number of scripts with odd scores remains unchanged or decreases by two. Consequently, if this process is repeated few times, then the number of scripts with odd scores would remain unchanged or decrease by an even number. Note that the box contains $4032 \times \frac{1}{2} = 2016$ scripts with odd scores. So after following the process certain number of times, the number of scripts with odd number of marks will always be an even number. The number of scripts with odd scores among the three left being at least one, is equal two. ■

Exercise 23. Consider the sequence of integers

$$1, 1, 2, 3, 5, 8, 13, 21, 34, \ldots,$$

where the first two terms are 1, and each subsequent term is the sum of the previous two terms. Prove that for any positive integer n, the greatest common divisor of the n-th and $(n+1)$-st terms is 1.

Walkthrough — Show that the greatest common divisor of the n-th and $(n+1)$-st terms is an invariant of the sequence.

Solution 22. For a positive integer n, let a_n denote the n-th term of the sequence. For any positive integer k, note that

$$\begin{aligned} \gcd(a_{k+2}, a_{k+1}) &= \gcd(a_{k+1} + a_k, a_{k+1}) \\ &= \gcd(a_k, a_{k+1}). \end{aligned}$$

By the principle of mathematical induction, it follows that for any positive integer n,

$$\gcd(a_n, a_{n+1}) = \gcd(a_1, a_2) = \gcd(1, 1) = 1.$$

■

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